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Semi-supervised GNN for Multi-modal Skin Lesion Classification

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Abstract

Accurate and trustworthy classification of skin lesions is critical for early melanoma screening. While deep learning models have shown strong performance, they often underutilize clinical metadata, provide limited reliability cues, and face challenges under limited labeled data. This paper presents a trust-aware semi-supervised Graph Neural Network (GNN) framework for multi-modal skin lesion classification with three components: (1) a metadata-aware attention fusion mechanism integrating deep visual features, handcrafted descriptors, and clinical metadata; (2) entropy-based uncertainty estimation for trust calibration; and (3) a selective pseudo-labeling strategy that augments training using only high-trust unlabeled samples. Given the severe class imbalance of ISIC 2020, we emphasize clinically meaningful and class-balanced evaluation (e.g., Sensitivity, Specificity, Macro F1, and Macro AUC) rather than accuracy alone. Across repeated stratified cross-validation, we observe that multimodal metadata fusion does not necessarily maximize purely supervised Phase 1 performance, but provides more stable training dynamics and representations that support effective trust-aware refinement. In Phase 2, entropy-filtered pseudo-labeling yields consistent improvements in macro-level metrics for the selected backbone, indicating enhanced minority-class discrimination when high-trust samples are used. The proposed framework further supports reliability-oriented analysis by reporting trust-level performance and trust-conditioned confusion matrices, providing practical transparency on where predictions are more or less reliable. Limitations include incomplete clinical metadata and the use of entropy as a lightweight (but not fully optimal) uncertainty proxy under extreme imbalance; these aspects motivate future exploration of stronger calibration and transformer-based multimodal fusion.

Keywords Skin lesion classification · GNN · Metadata fusion · Trust calibration · Pseudo-labeling · Explainable AI · Semi-supervised learning

1 Introduction

Skin cancer remains one of the most prevalent malignancies worldwide, and early detection plays a crucial role in reducing mortality. Recent advances in deep learning have led to substantial progress in automatic skin lesion classification, particularly through convolutional neural networks (CNNs), which have demonstrated strong capability in capturing discriminative visual patterns from dermoscopic images [1–6]. Despite these achievements,



many existing systems predominantly rely on image-only representations and often overlook important contextual cues such as patient demographics or lesion anatomical sites [7, 8]. A growing body of work has highlighted the complementary value of clinical metadata, which can enhance decision-making when effectively combined with visual evidence [9, 10]. Likewise, handcrafted descriptors—including color, texture, and shape features—have been explored as robust supplements to deep CNN features [7, 11].

Recent discussions in medical AI emphasize that, beyond improving predictive accuracy, the trustworthiness and interpretability of decision-support systems are key determinants in real-world clinical adoption [12, 13]. Explainable artificial intelligence (XAI) has emerged as an essential tool to provide clinicians with intuitive visual or semantic explanations of model reasoning [14–17]. Alongside interpretability, the ability to quantify predictive uncertainty has become increasingly important for identifying unreliable outputs that may require human intervention [18, 19]. Entropy-based confidence estimation has been shown to provide effective trust boundaries by categorizing predictions into high-, medium-, and low-confidence groups [18, 20]. Such trust calibration mechanisms offer a principled strategy for filtering out potentially erroneous predictions in clinical decision pipelines.

Another major challenge in medical imaging is the scarcity of labeled data. Semi-supervised learning and pseudo-labeling have therefore attracted increasing interest, enabling models to exploit unlabeled images by selectively incorporating high-confidence predictions during retraining [21–24]. The effectiveness of these methods strongly depends on robust confidence estimation and reliable selection of pseudo-labeled samples.

Graph Neural Networks (GNNs) have recently been investigated for skin lesion analysis due to their ability to model pairwise relationships among samples and capture higher-order structural patterns through graph propagation [25, 26]. By encoding sample affinities in a graph structure, GNNs allow global contextual reasoning that can complement local CNN-based feature extraction.

However, existing studies often investigate multimodal fusion, uncertainty estimation, graph-based learning, or pseudo-labeling as isolated components. In particular, limited attention has been paid to disentangling the individual contributions of (i) multimodal feature fusion, (ii) graph-based contextual modeling, and (iii) confidence-guided pseudo-labeling under severe class imbalance, which remains a critical concern in dermatological diagnosis. Moreover, high overall accuracy alone may be misleading in imbalanced medical datasets, motivating the use of class-balanced and clinically relevant metrics such as macro-averaged F1 score, sensitivity, specificity, and AUC.

In this work, we introduce a unified trust-aware semi-supervised framework that systematically integrates multimodal feature fusion, entropy-based trust calibration, and graph neural network modeling for skin lesion classification. The proposed approach is explicitly designed to address the limitations identified by recent critiques of medical AI systems, including overreliance on accuracy, lack of transparency, and insufficient analysis of uncertainty and minority-class behavior.

Our framework operates in two clearly separated phases. In Phase 1, a supervised GNN is trained on labeled data using multimodal node representations obtained via metadata-aware attention fusion. This phase enables a controlled comparison among multiple GNN architectures and isolates the contribution of graph-based learning from semi-supervised effects. In Phase 2, entropy-based trust estimation is applied to unlabeled samples to identify high-confidence predictions, which are selectively incorporated as pseudo-labels for self-retraining. This design allows us to explicitly analyze the impact of pseudo-labeling on class-balanced performance and model fairness.

To enhance interpretability and trustworthiness, the proposed system provides built-in trust-level analysis, including entropy distributions, accuracy-by-trust evaluation, and trust-conditioned confusion matrices, enabling a more transparent understanding of when and why the model is reliable. Rather than claiming explainability solely through predictive performance, the framework emphasizes uncertainty-aware reasoning and reliability assessment as core components of explainable clinical AI.

Extensive experiments conducted on the ISIC 2020 dataset demonstrate that the proposed framework consistently outperforms strong machine learning and GNN baselines in clinically meaningful metrics, while providing a principled mechanism to balance performance gains against reliability and minority-class sensitivity.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the proposed methodology, including multimodal fusion, graph construction, trust calibration, and pseudo-labeling. Section 4 describes the experimental setup and reports comprehensive results. Section 5 concludes the paper with final remarks and future directions.

2 Related Work

Deep learning has become the dominant paradigm for automatic skin lesion classification, largely due to its strong capability in extracting discriminative visual patterns from dermoscopic images. Numerous CNN-based models have been explored for melanoma detection and dermatological image understanding. For example, [1] introduced a multi-ResNet transformer hybrid model for atopic dermatitis diagnosis, achieving notable improvements in classification accuracy, while [2] proposed a clustering-augmented deep framework to enhance lesion subtype differentiation. Although these approaches demonstrated promising performance, they primarily relied on image features and did not incorporate clinical metadata.

Handcrafted descriptors have also been integrated into skin lesion classification pipelines to complement deep visual features. Zhao et al. [27] combined multiple handcrafted and learned representations, obtaining an AUC of 92.4% on ISIC 2020. However, handcrafted features alone often struggle to encode complex, high-level visual semantics. Ensemble architectures combining CNNs with classical machine learning classifiers have additionally been explored [28], yet they typically lack mechanisms for uncertainty estimation, a key requirement for deployment in clinical environments [14].

Metadata fusion has emerged as an important direction in dermatological AI research. Several studies have shown that incorporating patient demographics or lesion anatomical site information can improve classification robustness and reduce ambiguous predictions [9, 10]. Despite these findings, metadata remain underutilized in most deep-learning pipelines, and existing work rarely investigates principled mechanisms for jointly integrating metadata with both deep and handcrafted image features in a unified representation space.

Beyond predictive performance, the reliability and trustworthiness of AI-based diagnostic systems have gained increasing attention in recent medical AI literature. Trustworthy machine learning frameworks emphasize the importance of uncertainty quantification, systematic calibration, and interpretable decision-making, particularly for high-stakes clinical applications [29]. Among uncertainty estimation techniques, entropy-based confidence measures have been widely adopted as a lightweight and model-agnostic indicator of prediction reliability [18, 19]. Such uncertainty-aware strategies are especially relevant in semi-supervised learning, where pseudo-labeling methods rely on accurate confidence estimation to avoid error amplification during self-training [21–24].

Pseudo-labeling and self-training have a long history in deep learning research. Notable examples include the Noisy Student framework [30], which demonstrated large-scale self-training effectiveness on ImageNet, as well as more recent applications in medical imaging and related domains [31–33]. While these methods have shown that pseudo-labeling can improve performance under limited supervision, they are primarily designed for image-only CNN architectures and do not explicitly address multimodal fusion, graph-based relational modeling, or trust-level interpretability in severely imbalanced medical datasets.

Explainable AI (XAI) techniques have also been explored in dermatological classification to enhance transparency. Sharma et al. [34] proposed post-hoc saliency-based visualization methods, while Meng et al. [35] employed gradient-based explanation techniques. However, these approaches typically treat explainability as a post-processing step and do not integrate uncertainty estimation, trust calibration, or confidence-aware decision analysis into the learning pipeline itself.

Graph Neural Networks (GNNs) have recently gained attention in medical imaging for their ability to model relationships among samples through graph structures [25, 26]. By propagating information over similarity graphs, GNNs enable contextual reasoning that complements local CNN-based feature extraction. Nevertheless, existing GNN-based studies in dermatology often focus on single-modality inputs or supervised settings and rarely explore the joint integration of multimodal fusion, uncertainty-aware learning, and semi-supervised refinement.

Recent multimodal learning studies, such as [36], have reported that incorporating clinical metadata can yield modest performance gains (typically 2–3%) on skin lesion benchmarks. However, these approaches do not consider trust-aware pseudo-labeling or graph-based contextual learning, leaving the interaction between metadata, uncertainty, and relational modeling largely unexplored.

In summary, although substantial progress has been made in skin lesion classification using deep learning, important limitations remain. Existing approaches often (i) rely heavily on image-only representations, (ii) lack integrated uncertainty-aware or trust-calibrated learning mechanisms, and (iii) provide limited insight into minority-class behavior under severe class imbalance. These gaps motivate the development of a unified framework that combines multimodal feature fusion, trust-aware semi-supervised learning, and graph neural network modeling, as proposed in this study.

To contextualize our contribution, we further provide a structured summary of representative ISIC 2020 studies and their reported performance, highlighting the continued need for unified multimodal, trust-aware, and clinically reliable learning frameworks. Table 1

3 Methodology

This section formally defines the problem setting, introduces key concepts, and describes the proposed framework in a systematic manner, including theoretical formulation, algorithmic steps, and architectural design.

3.1 Problem Formulation

In this study, the problem of semi-supervised multi-modal skin lesion classification is addressed. Let $\mathcal{D}_L = \{(x_i, y_i)\}_{i=1}^{N_L}$ denote the labeled dataset, where x_i represents the input feature vector of the i -th sample and $y_i \in \{0, 1\}$ denotes its binary class label. Similarly, $\mathcal{D}_U = \{x_j\}_{j=1}^{N_U}$ denotes the unlabeled dataset with $N_U \gg N_L$.

Each input x is composed of multiple modalities, including deep visual features, handcrafted descriptors, and clinical metadata, fused via a feature extraction and integration function $\Phi(\cdot)$. The fused feature is defined in Eq. (1).

Table 1 Summary of recent works related to the ISIC 2020 dataset

Study	Task	Method	Reported Performance
Gupta et al. (2023) [37]	Classification	Knowledge distillation	Accuracy: 87.3%
Alhudhaif et al. (2021) [38]	Segmentation + Classification	Deep CNN approach	Accuracy: 85.5%
Abdelmaksoud et al. (2023) [39]	Classification	Residual multilevel features	Accuracy: 89.4%
Patel et al. (2022) [40]	Classification	Deep neural networks	AUC: 91.1%
Vashisth et al. (2024) [41]	Classification	Transformer + CNN fusion	Accuracy: 90.2%
Choudhary et al. (2023) [42]	Segmentation	Mask-RCNN + GRU model	Accuracy: 88.0%
Ali et al. (2024) [43]	Classification	Multilevel unsupervised learning network	AUC: 91.5%
Rana et al. (2023) [44]	Classification	Stacked CNN architectures	Accuracy: 86.7%
Kaur et al. (2024) [45]	Classification	Transfer learning approach	Accuracy: 89.8%

$$z_i = \Phi(x_i), \quad (1)$$

where z_i is the resulting multi-modal embedding for sample i .

To formally describe the multimodal learning task, we define each raw input as a triplet $x_i = (x_i^{\text{deep}}, x_i^{\text{hand}}, x_i^{\text{meta}})$, where x_i^{deep} denotes deep CNN features, x_i^{hand} represents handcrafted descriptors, and x_i^{meta} corresponds to structured clinical metadata. The fusion function $\Phi(\cdot)$ integrates these heterogeneous components into a unified representation z_i .

The classification task is modeled as a node-level prediction over a constructed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where each node $v_i \in \mathcal{V}$ corresponds to a fused feature z_i . The edges $\mathcal{E}$ represent similarity relationships among samples. The GNN predicts the class probability $\hat{y}_i$ for node v_i . The GNN outputs predicted probabilities as shown in Eq. (2).

$$\hat{y}_i = f_\theta(\mathcal{G}, z_i), \quad (2)$$

where f_θ denotes the GNN model parameterized by θ , following the standard GNN paradigm [25].

To estimate the uncertainty of each prediction, the Shannon entropy [46] of the predicted probability distribution is computed as defined in Eq. (3).

$$H(\hat{y}_i) = - \sum_{c=1}^C \hat{y}_{i,c} \log(\hat{y}_{i,c}), \quad (3)$$

where C is the number of classes and $\hat{y}_{i,c}$ is the predicted probability for class c .

Definition 1 (Prediction Entropy) Given a predicted probability distribution $\hat{y}_i = (\hat{y}_{i,1}, \hat{y}_{i,2}, \dots, \hat{y}_{i,C})$ over C classes for a sample i , the prediction entropy $H(\hat{y}_i)$ is defined by Eq. (3). It quantifies the uncertainty associated with the model's output, where a higher entropy value indicates greater uncertainty in classification.

In addition to entropy, we define a trust-level mapping function $T(\cdot)$ that categorizes predictions into three discrete confidence groups:

$$T(\hat{y}_i) \in \{\text{High}, \text{Medium}, \text{Low}\}, \quad (4)$$

where trust levels are derived by thresholding $H(\hat{y}_i)$ using quantiles estimated from the prediction distribution. High-trust predictions correspond to the lowest-entropy samples, whereas high-entropy predictions fall into the low-trust category. This discretization enables selective pseudo-labeling and targeted reliability analysis.

Samples from $\mathcal{D}_U$ with entropy below a threshold τ are selected for pseudo-labeling. The pseudo-label $\tilde{y}_j$ for an unlabeled sample x_j is assigned as defined in Eq. (5).

$$\tilde{y}_j = \arg \max_c \hat{y}_{j,c}, \quad \text{if } H(\hat{y}_j) \leq \tau, \quad (5)$$

where τ is determined based on trust calibration strategies, inspired by entropy-filtered semi-supervised learning approaches [19, 47].

The overall training objective $\mathcal{L}_{\text{total}}$ is composed of three components: the supervised classification loss on labeled data $\mathcal{L}_{\text{sup}}$, the pseudo-label loss on confident unlabeled data $\mathcal{L}_{\text{pseudo}}$, and an entropy regularization term $\mathcal{L}_{\text{entropy}}$ to encourage confident predictions. This combined objective is formally defined in Eq. (6).

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sup}} + \lambda_1 \mathcal{L}_{\text{pseudo}} + \lambda_2 \mathcal{L}_{\text{entropy}}, \quad (6)$$

where λ_1 and λ_2 are balancing hyperparameters.

This problem formulation enables the GNN model to leverage both labeled and confident unlabeled samples, while incorporating uncertainty-aware pseudo-labeling and multi-modal feature integration.

Overall, this formulation casts skin lesion classification as a *multimodal node classification* problem, where each lesion sample becomes a node enriched with deep visual features, handcrafted descriptors, and metadata-derived attributes. Uncertainty modeling through entropy and trust-level discretization provides a principled foundation for semi-supervised refinement using pseudo-labels.

3.2 Feature Extraction

3.2.1 Deep and Handcrafted Features

In this work, deep feature representations are extracted from dermoscopic images using pretrained convolutional neural networks (CNNs), including VGG19 [48], MobileNet [49], and DenseNet121 [50]. These architectures have been widely utilized in medical imaging tasks for their ability to capture hierarchical and abstract visual patterns [51].

Specifically, intermediate feature vectors are obtained by removing the classification heads of the networks and retaining the outputs from the last convolutional layers. This strategy has been shown effective for transfer learning in skin lesion classification [52, 53].

In addition to deep features, handcrafted descriptors such as color histograms and texture-based fractal features are considered to enrich the representation space. The complementary role of handcrafted features and deep embeddings has been demonstrated in previous work [9].

All extracted features are normalized to unit vectors before being used in subsequent fusion and graph construction stages.

To provide a clearer multimodal representation, the dimensionality of each feature type is explicitly defined as follows: the VGG19 feature vector has 512 dimensions, MobileNet provides a 1024-dimensional embedding, and DenseNet121 outputs a 1024-dimensional representation. These deep features capture progressively abstract properties including color patterns, texture, and high-level shape cues.

Handcrafted descriptors are extracted in a standardized manner. Color histograms are computed in RGB or HSV space and concatenated into a fixed-length vector, while fractal-based texture descriptors encode self-similarity patterns across spatial scales. These handcrafted features complement deep CNN embeddings by providing stable low-level cues that can help differentiate visually similar lesions.

3.2.2 Metadata Encoding

In addition to visual features, clinical metadata such as patient's age, sex, and anatomical site were extracted. These attributes have been shown to significantly enhance classification performance when combined with image features [54–56].

The metadata were processed by one-hot encoding for categorical variables (sex, anatomical site) and standard scaling for continuous variables (age). This approach follows the practices adopted in recent studies, where metadata were integrated with dermoscopic images to provide complementary contextual information [36, 57].

By incorporating metadata, the model could exploit epidemiological patterns of skin cancer distribution, which are not directly observable from images. Therefore, metadata extraction is considered an essential step in building a robust multi-modal classification system.

To ensure consistency and reproducibility, metadata preprocessing follows a standardized procedure. Age is normalized using MinMaxScaler to map numerical values into a comparable range, while missing age values are imputed with the median age of the dataset. Categorical attributes such as sex and anatomical site are transformed using one-hot encoding, allowing the model to learn lesion-location patterns or sex-dependent melanoma prevalence. Missing categorical values are replaced with an “unknown” token to preserve sample usability during training.

Formally, for each sample i , metadata are represented as $m_i = [\text{age}_i, \text{sex}_i, \text{site}_i]$ after preprocessing. Deep-image features x_i and metadata vectors m_i are fused through a metadata-aware attention module. The attention mechanism learns adaptive weights α_x and α_m to balance the contribution of image and metadata pathways.

The image and metadata features are combined through a metadata-aware attention fusion module. Attention weights α_m and α_x are learned to dynamically balance metadata and image information. The fused representation h_i for sample i is computed as defined in Eq. (7):

$$h_i = \alpha_x \cdot x_i + \alpha_m \cdot m_i, \quad (7)$$

where x_i and m_i denote the image and metadata features, respectively.

This attention-based fusion scheme enables the model to emphasize metadata when it contributes strongly to discriminating benign from malignant lesions (e.g., age and anatomical site), while relying more heavily on deep or handcrafted image features when metadata provide limited contextual cues.

3.3 Graph Construction and GNN Modeling

In this study, a graph structure is constructed to capture the complex relationships among skin lesion samples. Each sample is represented as a node, and the edges between nodes are designed to encode the similarity between feature representations.

Definition 2 (Graph Definition) The graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ consists of a set of nodes $\mathcal{V}$, where each node $v_i \in \mathcal{V}$ corresponds to a skin lesion sample, and a set of edges $\mathcal{E}$ representing the connections between nodes.

Each node v_i is associated with a feature vector $\mathbf{x}_i \in \mathbb{R}^d$, extracted by concatenating deep features, handcrafted features, and processed metadata attributes. Deep features are obtained from convolutional neural networks such as VGG19 [48]. Handcrafted features include color histograms and fractal descriptors. Metadata features such as age, sex, and anatomical site are encoded by one-hot encoding and standardized scaling as commonly adopted in medical imaging tasks [52].

Edges are constructed based on a similarity function. Specifically, cosine similarity is employed to quantify the relationship between node features. It is formally defined in Eq. (8):

$$s(\mathbf{x}_i, \mathbf{x}_j) = \frac{\mathbf{x}_i^\top \mathbf{x}_j}{\|\mathbf{x}_i\|_2 \|\mathbf{x}_j\|_2}. \quad (8)$$

An undirected edge $(v_i, v_j) \in \mathcal{E}$ is created if $s(\mathbf{x}_i, \mathbf{x}_j)$ exceeds a predefined similarity threshold τ .

In addition to threshold-based edges, a k -nearest neighbor (k-NN) graph construction strategy is incorporated to ensure consistent graph connectivity. Empirically, $k = 5$ is chosen because it provides a balance between capturing meaningful local neighborhoods and avoiding excessive densification of the graph. Too small a value of k may lead to disconnected components that hinder message passing, whereas large values of k may introduce noisy or clinically irrelevant edges. Importantly, the k-NN graph is constructed without using any label information, and training/validation/test masks ensure that no data leakage occurs during model training.

Once the graph structure is established, a Graph Convolutional Network (GCN) [58] is adopted to learn latent node representations. In each GCN layer, the node features are updated based on their neighbors according to the propagation rule defined in Eq. (9).

$$\mathbf{H}^{(l+1)} = \sigma \left(\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{H}^{(l)} \mathbf{W}^{(l)} \right), \quad (9)$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ is the adjacency matrix with self-loops, $\tilde{\mathbf{D}}$ is its corresponding degree matrix, $\mathbf{W}^{(l)}$ are trainable weights at layer l , and $\sigma(\cdot)$ is a non-linear activation function, typically ReLU.

Beyond the standard GCN layer, several GNN architectures are supported in the proposed framework to enrich expressive capacity. Graph Attention Networks (GAT) apply learnable attention coefficients to weight neighboring nodes adaptively. GraphSAGE aggregates information via mean or pooling operators and enables inductive generalization to unseen nodes. The GINE architecture introduces stronger non-linear message transformations suitable for heterogeneous multimodal features. A hybrid GraphSAGE–GAT variant is also examined to combine the benefits of pooling-based and attention-based aggregation. Additionally, the GCN implementation follows a residual design in the codebase to mitigate oversmoothing effects and improve gradient flow in deeper layers.

The training of the GNN is conducted by minimizing a supervised cross-entropy loss over the labeled nodes $\mathcal{V}_L$, formally defined in Eq. (10).

$$\mathcal{L}_{\text{sup}} = \frac{1}{|\mathcal{V}_L|} \sum_{v_i \in \mathcal{V}_L} \mathcal{L}_{\text{CE}}(f(h_i), y_i), \quad (10)$$

where $f(\cdot)$ denotes the final classifier head mapping the node embedding h_i to the predicted label.

In order to leverage the information from the unlabeled nodes $\mathcal{V}_U$, pseudo-labeling is performed based on entropy confidence filtering [47]. High-confidence predictions are selected to construct pseudo-labels, leading to an auxiliary loss term that encourages consistent predictions on confident samples. This pseudo-labeling loss is formally defined in Eq. (11):

$$\mathcal{L}_{\text{pseudo}} = \frac{1}{|\mathcal{V}_U^{\text{conf}}|} \sum_{v_i \in \mathcal{V}_U^{\text{conf}}} \mathcal{L}_{\text{CE}}(f(h_i), \hat{y}_i), \quad (11)$$

where $\mathcal{V}_U^{\text{conf}}$ denotes the confident unlabeled nodes, and $\hat{y}_i$ are their corresponding pseudo-labels.

Prior to graph propagation, multimodal node embeddings are further enhanced through a MetadataAttentionFusion module. This module computes query (Q), key (K), and value (V) projections from metadata and image features, enabling metadata to reweight the contribution of visual embeddings via an attention mechanism. Residual connections and layer normalization stabilize the fusion process, preventing metadata from being overshadowed by high-dimensional image features and ensuring improved interpretability in multimodal integration.

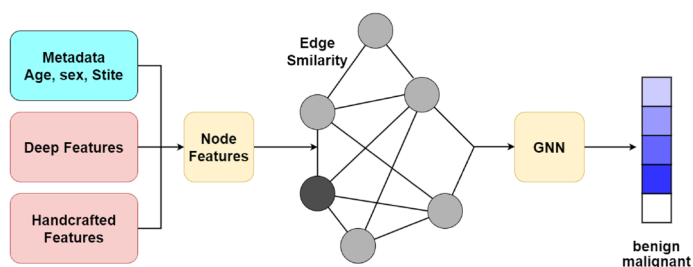
The training process thus propagates features through the graph structure, aggregates information from similar samples, and adapts to unlabeled data using trust-aware pseudo-labeling, leading to a robust and interpretable skin lesion classification model.

In our framework, the construction of the graph and the subsequent GNN modeling process are divided into two distinct phases, as illustrated in Fig. 1.

In the first phase, feature extraction is performed to represent each skin lesion sample. Deep features are extracted from pretrained convolutional neural networks (CNNs) such as VGG19 [48]. Handcrafted features including color histograms, texture descriptors, and fractal analysis are computed to provide complementary information. Clinical metadata such as patient age, sex, and anatomical site are processed and normalized. All feature modalities are concatenated to form comprehensive node features.

In the second phase, a graph is constructed where each node represents a lesion sample characterized by its extracted feature vector. Edges between nodes are established based on feature similarity, typically measured by cosine similarity or Euclidean distance. A Graph Neural Network (GNN) [58] is then trained on this constructed

Fig. 1 The overall pipeline for graph construction and GNN training. Phase 1: Feature extraction for node representation. Phase 2: Graph construction and node classification via GNN



graph to predict the class label of each lesion node (i.e., benign or malignant). During training, supervised loss is minimized over labeled nodes, while semi-supervised learning is enabled through graph propagation.

After constructing the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ and defining the node features as discussed, an initial supervised training of the GNN model is conducted. This training phase aims to optimize the model parameters θ by minimizing the supervised loss $\mathcal{L}_{\text{sup}}$ (Eq. (10)) over the labeled nodes $\mathcal{V}_L$.

The goal is to obtain a well-trained initial model θ^* that can serve as the base for subsequent trust calibration and pseudo-labeling. The procedure of initial GNN training is summarized in Algorithm 1.

```

1: Construct graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$  from extracted features.
2: Initialize GNN model  $f_\theta$  with random weights.
3: for epoch = 1 to  $T_{\text{max}}$  do
4:   Forward pass: compute node embeddings  $\mathbf{H}$  and predictions  $\hat{y}_i$  using  $f_\theta$ .
5:   Compute supervised loss  $\mathcal{L}_{\text{sup}}$  according to Equation (10).
6:   Update parameters  $\theta$  by backpropagation.
7:   Evaluate model on validation set.
8:   if current validation performance is best then
9:     Save model checkpoint  $\theta^*$ .
10:  end if
11: end for
12: Return the best model parameters  $\theta^*$ .

```

Algorithm 1 Initial GNN Training Algorithm

3.4 Trust Calibration and Pseudo-labeling

In semi-supervised graph-based learning, it is essential to identify and utilize high-confidence predictions to augment the labeled training set. To achieve this, predictive uncertainty is estimated through the entropy of the output probability distribution, as defined in Definition 2 and Eq. (3).

Entropy serves as a widely used uncertainty measure in both binary and multiclass classification tasks. For binary classification, entropy reduces to a simple function of the predicted positive probability and reaches its maximum when the model is maximally uncertain (i.e., $p = 0.5$). For multiclass settings, entropy naturally extends to capture distributional ambiguity when probability mass is spread across classes. This makes entropy an appropriate and model-agnostic confidence indicator for trust calibration.

Based on the entropy, trust calibration is performed to filter samples with sufficiently low uncertainty. Specifically, given an unlabeled sample $\mathbf{x}_j \in \mathcal{D}_U$, its predicted class probabilities $\hat{y}_j$ are obtained by the trained GNN model following Eq. (2). The entropy $H(\hat{y}_j)$ is computed, and pseudo-labels $\tilde{y}_j$ are assigned according to Eq. (5) if the confidence criterion $H(\hat{y}_j) \leq \tau$ is satisfied.

The construction of the pseudo-labeled set $\mathcal{D}_P$ can be formalized as a subset of $\mathcal{D}_U$ containing only those samples whose predicted entropy falls below the confidence threshold τ . This selection rule is mathematically defined in Eq. (12).

$$\mathcal{D}_P = \{(\mathbf{x}_j, \tilde{y}_j) \mid \mathbf{x}_j \in \mathcal{D}_U, H(\hat{y}_j) \leq \tau\}. \quad (12)$$

To better interpret entropy values in practice, the predicted samples are divided into three trust levels using quantile-based segmentation: High Trust (entropy in the bottom 0.33 quantile), Medium Trust (0.33–0.66), and Low Trust (top 0.66 quantile). This division reflects natural clustering patterns in model confidence and is widely used to approximate reliability distributions in medical AI. Empirically, accuracy is found to correlate inversely with entropy, with High Trust samples exhibiting significantly higher prediction correctness. Such segmentation allows more interpretable monitoring of model reliability and supports selective pseudo-labeling.

After pseudo-labeling, the GNN is retrained by minimizing the total loss $\mathcal{L}_{\text{total}}$ as formulated in Eq. (6), where $\mathcal{L}_{\text{sup}}$ is the supervised cross-entropy loss on $\mathcal{D}_L$ (Eq. (10)), $\mathcal{L}_{\text{pseudo}}$ is the pseudo-labeling loss on $\mathcal{D}_P$ (Eq. (11)), and $\mathcal{L}_{\text{entropy}}$ is an entropy regularization term to encourage confident predictions.

Only High Trust samples are used for pseudo-labeling to ensure strong reliability of added labels and to avoid error amplification during retraining. A minimum number of pseudo-labeled samples is required to trigger Phase 2 (pseudo-retraining). If the number of High Trust samples is insufficient, retraining is skipped to prevent degradation of model performance. This design ensures that Phase 2 behaves as a confidence-driven refinement stage rather than an unconditional training cycle.

After obtaining the best model parameters θ^* from the initial GNN training phase (Algorithm 1), a trust-aware pseudo-labeling strategy is applied to leverage the unlabeled dataset $\mathcal{D}_U$.

The model f_{θ^*} is utilized to generate soft predictions for each unlabeled sample. Based on the prediction entropy (Definition 1, Eq. (3)), confident samples are selected and pseudo-labeled according to the criterion defined in Eq. (5).

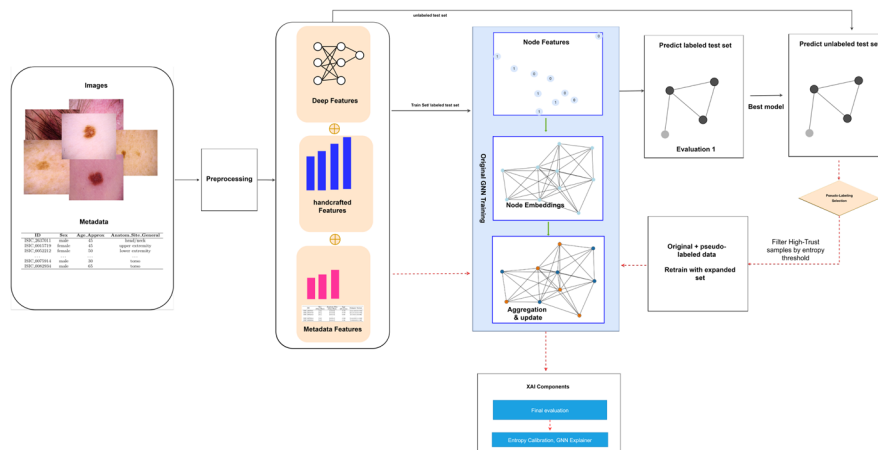
The complete procedure of this trust calibration and pseudo-labeling framework is summarized in Algorithm 2.

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- 1: **Input:** Labeled set $\mathcal{D}_L$, unlabeled set $\mathcal{D}_U$, threshold τ
 - 2: Train GNN f_{θ} on $\mathcal{D}_L$ by minimizing $\mathcal{L}_{\text{sup}}$ (Equation (10)).
 - 3: **for** each unlabeled sample $\mathbf{x}_j \in \mathcal{D}_U$ **do**
 - 4: Predict $\hat{y}_j = f_{\theta^*}(\mathcal{G}, \mathbf{z}_j)$ for each unlabeled sample $\mathbf{z}_j$ according to Equation (2).
Here θ^* denotes the best model parameters obtained from Algorithm 1.
 - 5: Compute entropy $H(\hat{y}_j)$ using Equation (3).
 - 6: **if** $H(\hat{y}_j) \leq \tau$ **then**
 - 7: Assign pseudo-label $\tilde{y}_j$ as in Equation (5).
 - 8: Add $(\mathbf{x}_j, \tilde{y}_j)$ to $\mathcal{D}_P$ as in Equation (12)
 - 9: **end if**
 - 10: **end for**
 - 11: Retrain GNN f_{θ} on the expanded set $\mathcal{D}_L \cup \mathcal{D}_P$ by minimizing $\mathcal{L}_{\text{total}}$ (Equation (6)).
-

Algorithm 2 Trust-Aware Pseudo-Labeling Framework

This trust-aware semi-supervised learning framework ensures that only highly confident samples are incorporated into the retraining process, thereby mitigating the risk of error propagation from noisy pseudo-labels. The entropy-based filtering strategy follows principles from recent studies [19, 47], and its integration into graph neural networks enables robust training in medical imaging contexts.

Fig. 2 Overview of the proposed trust-aware semi-supervised multi-modal GNN framework. Phase 1: Initial supervised GNN training on labeled data (Algorithm 1). Phase 2: Trust-aware pseudo-labeling and retraining with confident unlabeled samples (Algorithm 2)



3.5 Overall System Architecture

The overall workflow of the proposed Trust-Aware Semi-Supervised Multi-Modal GNN framework is organized into two main phases, as depicted in Fig. 2. This two-stage structure enables effective exploitation of both labeled and unlabeled data, while ensuring model reliability through entropy-based trust calibration.

The architecture explicitly integrates four major components: (i) multimodal attention fusion to combine deep features, handcrafted descriptors, and clinical metadata, (ii) graph construction based on multimodal similarity, (iii) supervised GNN learning in Phase 1, and (iv) trust-aware pseudo-labeling and self-retraining in Phase 2. These components operate sequentially as a unified pipeline.

In Phase 1, multi-modal feature extraction is performed. Each skin lesion sample is represented by fusing deep visual features extracted from pretrained CNNs (such as VGG19 [48]), handcrafted features (color histograms, fractal descriptors), and clinical metadata (age, sex, anatomical site). These feature modalities are integrated through a metadata-aware attention mechanism, resulting in comprehensive node representations. Based on these fused features, a similarity-based graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is constructed. An initial supervised training of the GNN model is then conducted using only the labeled set $\mathcal{D}_L$, following Algorithm 1, in order to obtain the best model parameters θ^* .

Phase 1 therefore encapsulates (a) multimodal attention fusion, (b) feature normalization, (c) similarity-based or k-NN graph construction, and (d) supervised GNN optimization. This phase produces a stable baseline model suitable for reliable uncertainty estimation.

In Phase 2, trust calibration and semi-supervised learning are applied. The trained model f_{θ^*} predicts soft labels for samples in the unlabeled set $\mathcal{D}_U$. Prediction uncertainty is quantified via entropy (Definition 1, Eq. (3)). Samples with entropy below a threshold τ are selected as high-confidence predictions. Pseudo-labels are assigned and the pseudo-labeled set $\mathcal{D}_P$ is created. The GNN is then retrained on the expanded dataset $\mathcal{D}_L \cup \mathcal{D}_P$ following the procedure detailed in Algorithm 2.

Phase 2 introduces a trust-aware mechanism in which pseudo-labels are filtered exclusively from High Trust samples. This mechanism mitigates error propagation and stabilizes semi-supervised learning. The refined model captures a broader decision boundary informed by both labeled and reliable unlabeled samples.

The clear separation between initial supervised training and entropy-filtered pseudo-labeling ensures robustness and prevents noise propagation from low-confidence predictions.

An updated conceptual diagram is illustrated in Fig. 2, emphasizing the two-stage design (Phase 1 supervised learning and Phase 2 trust-aware pseudo-labeling), and explicitly showing the multimodal attention fusion block as well as the graph construction module.

3.6 Main Contributions

The main contributions of this article are summarized as follows:

1. A trust-aware semi-supervised GNN framework is proposed, where deep features, handcrafted descriptors, and clinical metadata are fused through a metadata-aware attention mechanism.
2. An entropy-based trust calibration strategy is introduced to segment predictions into high, medium, and low trust levels for pseudo-label selection.
3. A self-retraining process based on high-trust pseudo-labels is implemented to enhance model performance without requiring additional manual annotations.
4. An end-to-end explainable pipeline is developed, enabling the visualization of entropy distribution, trust-based confusion matrices, and metadata attention patterns.
5. Extensive experiments are conducted on the ISIC 2020 dataset, showing consistent improvements over traditional machine learning and deep learning baselines.

In addition, the proposed framework incorporates a multimodal fusion mechanism based on metadata-aware attention, enabling effective integration of deep, handcrafted, and clinical features. The system further employs a structured trust-aware pseudo-labeling process, complemented by a comprehensive cross-validation protocol to ensure robust evaluation. Sensitivity and specificity analyses are included to highlight clinical relevance, while the relationship between trust levels and predictive accuracy is examined to provide deeper insights into model reliability.

4 Experiments

This section presents the experimental framework used to evaluate the proposed trust-aware semi-supervised multi-modal GNN model, including dataset characteristics, preprocessing steps, training configuration, baseline comparisons, and evaluation metrics.

4.1 Dataset and Preprocessing

In this study, the ISIC 2020 Challenge dataset [59] was utilized to evaluate the proposed method. The dataset consists of 33,126 dermoscopic images accompanied by patient-level clinical metadata, including approximate age, sex, and anatomical site of the lesion. Each image is annotated for binary classification into benign or malignant categories.

4.1.1 Metadata Missing Statistics

A quantitative analysis of metadata completeness was performed separately for the training and test splits of the ISIC 2020 dataset. Missing-value statistics were computed directly from the official metadata files and are summarized in Table 2. In the training set, missing values account for 7.3% of the sex attribute, 10.5% of the anatomical site labels, and 2.4% of age information. The test set exhibits a comparable missingness pattern, with slightly higher absence rates for anatomical site and age. All statistics were calculated independently for each data split to ensure a leakage-free preprocessing procedure.

Missing categorical attributes (sex and anatomical site) were mapped to a dedicated `unknown` category. Missing continuous values (age) were imputed using the mean age computed exclusively from the training subset. The same training-derived statistics were subsequently applied to the validation and test sets, ensuring strict prevention of information leakage.

4.1.2 Class Distribution Visualization

To visualize the inherent class imbalance of the dataset, the true class distribution after ground-truth label alignment was examined. Figure 3 and Table 3 show that the ISIC 2020 dataset is severely imbalanced, containing 32,542 benign samples and only 584 malignant cases ($\approx 1.8\%$). Importantly, no data-level class balancing or augmentation techniques were applied in this study. Instead, this pronounced imbalance motivates the adoption of trust-aware semi-supervised learning and entropy-based pseudo-labeling strategies.

Table 2 Missing metadata statistics for the ISIC 2020 dataset

Metadata field	Training set (%)	Test set (%)
Sex	7.3	7.1
Anatomical site	10.5	11.2
Age	2.4	2.6

Fig. 3 True benign versus malignant class distribution in the ISIC 2020 dataset after ground-truth label alignment. The dataset exhibits a severe class imbalance (32,542 benign vs. 584 malignant samples). No data-level class balancing or augmentation is applied

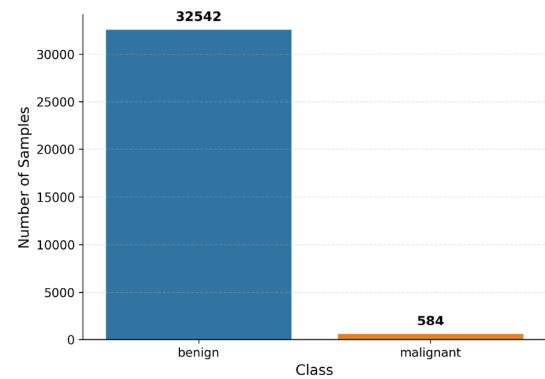


Table 3 Class Distribution after Ground-Truth Label Alignment

Class	Number of Samples	Percentage (%)
Benign	32,542	98.2
Malignant	584	1.8

Table 4 Representative examples of raw clinical metadata from the ISIC 2020 dataset

Image ID	Sex	Age (Approx.)	Anatomical Site
ISIC_2637011	Male	45	Head/neck
ISIC_0015719	Female	45	Upper extremity
ISIC_0052212	Female	50	Lower extremity
ISIC_0075914	Male	30	Torso
ISIC_0082934	Male	65	Torso

Table 5 Encoded clinical metadata after normalization and one-hot encoding

Image ID	Sex (One-Hot)	Site (One-Hot)	Age (Z-score)	Encoded Feature Vector
ISIC_2637011	[1,0]	[1,0,0,0]	-0.50	[1,0,1,0,0,0,-0.50]
ISIC_0015719	[0,1]	[0,1,0,0]	-0.50	[0,1,0,1,0,0,-0.50]
ISIC_0052212	[0,1]	[0,0,1,0]	0.00	[0,1,0,0,1,0,0.00]
ISIC_0075914	[1,0]	[0,0,0,1]	-2.00	[1,0,0,0,0,1,-2.00]
ISIC_0082934	[1,0]	[0,0,0,1]	1.50	[1,0,0,0,0,1,1.50]

All dermoscopic images were resized to a uniform resolution prior to feature extraction. Deep feature representations were obtained using pretrained convolutional neural networks, namely VGG19 [48], MobileNet [49], and DenseNet121 [50]. The classification layers of these networks were removed, and activations from the final convolutional layers were retained as feature vectors. All deep features were normalized to unit norm to ensure numerical consistency across modalities.

In addition to deep features, handcrafted descriptors were extracted to enrich the representation space. Specifically, color histograms, HSV histograms, and fractal features were employed. Feature relevance was assessed using Chi-square statistical analysis following [60]. The most discriminative handcrafted features were concatenated with deep features to form a comprehensive visual embedding.

Clinical metadata were encoded by applying one-hot encoding to categorical variables (sex and anatomical site), while the age attribute was standardized using Z-score normalization.

4.1.3 Metadata Normalization

Clinical metadata were normalized using a leakage-safe preprocessing protocol to ensure consistency and fair evaluation across data splits. For the continuous age attribute, normalization parameters (mean and standard deviation) were estimated exclusively from the training subset and subsequently applied to the validation and test

splits. Categorical variables, including sex and anatomical site, were transformed using one-hot encoding with an explicit `unknown` category to handle missing or unspecified values. This strategy guarantees fixed-dimensional metadata representations across all splits, thereby preserving feature alignment during multimodal fusion.

Representative examples of raw clinical metadata and their corresponding encoded feature vectors are presented in Tables 4 and 5. The resulting normalized metadata vectors were concatenated with deep and handcrafted visual features to form unified multimodal embeddings for each sample, which serve as node attributes in the subsequent graph-based learning framework.

Feature selection was guided by Chi-square ranking computed on the training set. Figure 4 illustrates the average Chi-square scores for each feature type, indicating that DenseNet-based deep features and color-based handcrafted descriptors provide the strongest discriminative power.

Based on this analysis, only the most informative feature modalities were selected for subsequent graph construction and trust-aware GNN modeling. All selected visual features and encoded clinical metadata were concatenated to form a unified multimodal representation for each sample, enabling robust semi-supervised graph-based classification under severe class imbalance without relying on explicit data-level rebalancing techniques.

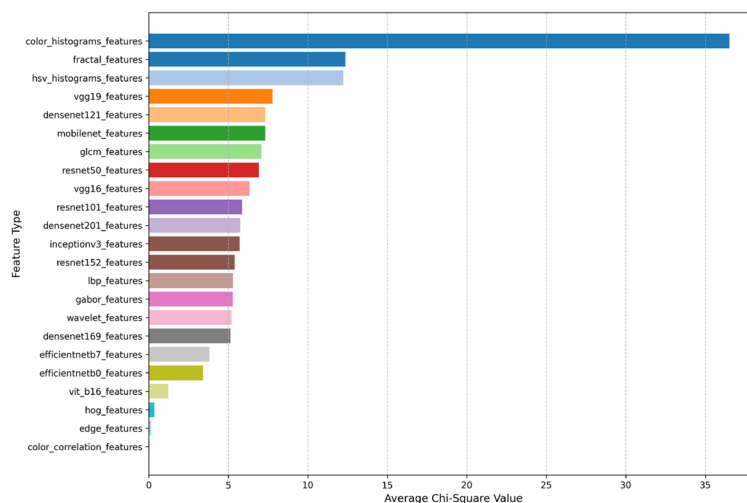
4.2 Experimental Setup

All experiments were conducted on the ISIC 2020 dataset. We adopt repeated stratified k -fold cross-validation to obtain robust estimates under severe class imbalance. Specifically, we use $k = 5$ folds and repeat the procedure $n = 3$ times, resulting in 15 fold evaluations. The implementation follows `RepeatedStratifiedKFold` to preserve class proportions across splits and reduce split-to-split variance.

Input features and preprocessing. Each sample is represented by a multimodal feature vector composed of deep CNN embeddings (VGG19, MobileNet, DenseNet121), handcrafted descriptors (color/HSV histograms and fractal features), and encoded clinical metadata (sex, age, anatomical site). Deep and handcrafted features are extracted offline as described in Sect. 4.1. Clinical metadata are normalized using leakage-safe statistics computed from the training split only (Sect. 4.1). All feature vectors are ℓ_2 -normalized before graph construction.

Graph construction. For each fold, we build a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where each node corresponds to a lesion sample. Edges are created using a k -nearest neighbor (k-NN) strategy based on cosine similarity over the multimodal node features. Unless otherwise stated, we set $k = 5$ and symmetrize the adjacency to form an undirected graph. Graph construction is label-free; training and validation supervision is applied only through fold-specific node masks, ensuring no label leakage.

Fig. 4 Average Chi-square ranking of feature types on the ISIC 2020 dataset



GNN backbones and training. We evaluate multiple GNN architectures under identical settings: GCN [58], GAT [61], GraphSAGE [62], GINE [63], and a hybrid GraphSAGE–GAT model. All backbones use two message-passing layers with hidden dimension 128, ReLU activations, and dropout rate 0.5. Models are trained with Adam [64] using an initial learning rate of 10^{-3} . A ReduceLROnPlateau scheduler halves the learning rate if the validation loss does not improve for 5 epochs, and early stopping is applied with a patience of 10 epochs. The maximum number of epochs is 150 and the batch size is 64.

Phase 2 trust-aware pseudo-label retraining. After Phase 1, the selected backbone (GINE) is further refined via entropy-based trust calibration and pseudo-labeling, yielding `GINE_pseudo_finetune`. High-confidence predictions are filtered by entropy and used as pseudo-labels for self-retraining, while the evaluation protocol remains fold-consistent and label-free during pseudo-label generation.

Reproducibility and implementation. Each experiment is repeated with three random seeds. We report class-balanced and clinically meaningful metrics (Sensitivity, Specificity, Macro F1, Macro AUC) as mean $\pm$ standard deviation across the repeated stratified folds. All experiments are implemented in Python using PyTorch and PyTorch Geometric, and trained on a single NVIDIA RTX 3090 (24GB).

4.3 Baselines and Comparison Methods

To assess the effectiveness of the proposed trust-aware semi-supervised multi-modal GNN framework, four baseline scenarios were designed to isolate the contributions of different feature modalities and graph-learning components. The scenarios were constructed to ensure fair comparison between traditional machine learning methods, established GNN architectures, and the proposed model.

Scenario 1 (Machine Learning with All Features). This scenario serves as a non-graph baseline for comparison with the proposed GNN-based framework. All available modalities, including deep features (VGG19, MobileNet, DenseNet121), handcrafted descriptors (color histograms, HSV histograms, and fractal features), and clinical metadata (sex, age, and anatomical site), were concatenated into a single feature vector. Four representative classical models were evaluated: Logistic Regression, Support Vector Machine, Random Forest, and XGBoost. These baselines cover linear, margin-based, bagging, and boosting paradigms, providing a compact yet comprehensive reference to assess the benefits of graph modeling and trust-aware learning.

Scenario 2 (GNN Baselines without Metadata). This scenario examines the ability of graph neural networks to learn from image-derived features alone. Only deep and handcrafted visual features were included; all metadata fields were removed. The following GNN architectures were evaluated: GCN [58], GAT [61], GraphSAGE [62], GINE [63], and a hybrid GraphSAGE–GAT model. These correspond to the implemented list `GNN_MODELS_PHASE1 = {GCN, GAT, GraphSAGE, GINE, GraphSAGEGAT}`.

Scenario 3 (GNN Baselines with All Features). This scenario extends Scenario 2 by incorporating clinical metadata into the graph input. Deep features, handcrafted descriptors, and metadata were fused to create multi-modal node embeddings. This configuration enables direct comparison with Scenario 2, thereby quantifying the benefit of metadata integration for graph-based learning.

Scenario 4 (Proposed Model – Trust-Aware Semi-Supervised Learning). This scenario evaluates the complete proposed framework described in Sect. 3, which integrates multimodal feature fusion, graph-based learning, and uncertainty-aware semi-supervised training.

First, node representations were constructed by fusing deep image features, handcrafted descriptors, and clinical metadata through a metadata-aware attention fusion module. Based on the fused representations, a k -nearest neighbor graph was built, and an initial supervised GraphSAGE model was trained using labeled samples only (**Phase 1 – GraphSAGE**).

Next, entropy-based uncertainty estimation was applied to the unlabeled test set to quantify prediction confidence. Samples were partitioned into High, Medium, and Low trust levels according to entropy quantiles. Only High-trust predictions were retained to generate pseudo-labels, which were then incorporated into a self-retraining process (**Phase 2 – GraphSAGE_pseudo_finetune**).

This trust-guided pseudo-labeling strategy ensures that only reliable unlabeled samples contribute to model refinement, thereby reducing error accumulation during self-training. The resulting two-phase learning scheme enables the model to leverage unlabeled data while preserving predictive reliability.

All models in Scenarios 1–3 were trained under identical experimental settings using repeated stratified cross-validation. Hyperparameters were tuned independently for each baseline. Model performance was evaluated using accuracy, F1-score, sensitivity, specificity, and AUC.

Overall, this scenario represents the full proposed approach and enables a comprehensive comparison between (i) traditional machine learning and graph-based learning, (ii) GNN models with and without multimodal metadata fusion, and (iii) standard supervised GNN baselines and the proposed trust-aware semi-supervised framework.

4.4 Evaluation Metrics

To evaluate the performance of the proposed model and the baselines, several standard classification metrics were employed. These include Accuracy (ACC), Area Under the ROC Curve (AUC), Precision, Recall, and F1-score.

Accuracy measures the proportion of correctly classified samples and is commonly used for overall evaluation [65]. Precision quantifies the proportion of true positive predictions among all positive predictions, while Recall measures the proportion of true positive predictions among all actual positives [65]. F1-score, defined as the harmonic mean of Precision and Recall, is particularly important in imbalanced classification tasks such as skin lesion detection [66].

The AUC metric evaluates the ability of the model to discriminate between classes across different thresholds, and is especially recommended for medical diagnosis tasks where decision thresholds vary [67].

In addition to these metrics, Sensitivity (also referred to as True Positive Rate) and Specificity (True Negative Rate) were included due to their clinical significance. Sensitivity measures the model's ability to correctly detect malignant lesions, while Specificity reflects its ability to correctly identify benign lesions. Both metrics are crucial in binary dermatological classification, where failing to detect malignancy may lead to delayed diagnosis and adverse outcomes.

$$\text{Sensitivity} = \frac{TP}{TP + FN}, \quad \text{Specificity} = \frac{TN}{TN + FP}.$$

For multi-class extensions, macro-averaged Sensitivity and Specificity were also computed. These metrics treat each class equally by averaging per-class values, thereby avoiding dominance by majority classes in imbalanced settings.

$$\text{Macro-Sensitivity} = \frac{1}{C} \sum_{c=1}^C \frac{TP_c}{TP_c + FN_c}, \quad \text{Macro-Specificity} = \frac{1}{C} \sum_{c=1}^C \frac{TN_c}{TN_c + FP_c}.$$

To further quantify model reliability, trust-level evaluation was conducted by segmenting predictions into High-, Medium-, and Low-trust groups based on entropy thresholds defined in Sect. 3.4. For each trust level, accuracy was computed separately to generate “accuracy-by-trust” curves. These curves reveal how prediction confidence correlates with correctness, providing a more granular assessment of reliability beyond global accuracy.

Additionally, confusion matrices were computed for each trust segment. Trust-conditioned confusion matrices allow examination of how error types (e.g., false negatives in malignant cases) vary across trust levels. This stratified evaluation is essential in safety-critical medical applications, as it identifies which confidence regions contain most diagnostic risks.

The definitions and formulas of the metrics used in this study follow standard conventions in machine learning and biomedical classification literature [65–67], ensuring fair and reproducible comparisons.

4.5 Results and Analysis

This section presents a comprehensive evaluation of the proposed trust-aware semi-supervised multi-modal GNN framework across the four experimental scenarios defined in Sect. 4.3. Performance comparisons, metadata ablation, trust-level reliability analysis, and qualitative insights are provided through quantitative tables and visual figures.

4.5.1 Phase 1 Performance of GNN Models

This subsection reports the performance of supervised models in Phase 1, prior to trust calibration and pseudo-label retraining. Both traditional machine learning (ML) baselines and graph neural network (GNN) architectures were evaluated under identical repeated stratified cross-validation settings. Given the severe class imbalance of the ISIC 2020 dataset, performance assessment focuses on class-balanced and clinically relevant metrics rather than accuracy.

Evaluation metrics. To ensure a fair comparison between ML and GNN paradigms, we report Sensitivity, Specificity, Macro F1-score, and Macro AUC, each summarized as mean $\pm$ standard deviation across folds. Sensitivity reflects the ability to detect malignant lesions, while Macro F1 and Macro AUC provide class-balanced measures that are robust to skewed label distributions.

Quantitative comparison. As shown in Table 6, ML baselines achieve extremely high Specificity and Macro AUC values, which translate into excellent overall accuracy. However, their Sensitivity remains consistently low, particularly for Random Forest and SVM, indicating a strong bias toward the majority benign class. This behavior leads to a high false-negative rate for malignant lesions, which is undesirable in clinical screening scenarios.

In contrast, GNN-based models demonstrate substantially improved Sensitivity and more balanced Macro AUC scores. By leveraging relational structure and metadata-aware learning, GNNs reduce majority-class dominance and enhance malignant detection. Among the evaluated architectures, GINE and GraphSAGE–GAT provide the most favorable trade-off between Sensitivity and Macro AUC in Phase 1.

Confusion matrix analysis. To further investigate prediction behavior, Figs. 5 and 6 present the normalized confusion matrices of representative ML and GNN models, respectively.

Overall, the combined quantitative metrics and confusion matrix analysis demonstrate that Phase 1 GNN models provide a clinically more meaningful operating point than traditional ML baselines. These findings motivate the adoption of a GNN backbone for trust-aware pseudo-label selection and semi-supervised retraining in Phase 2.

Statistical significance analysis. To verify that the performance gains introduced by Phase 2 are not due to split-to-split variation, we conducted *paired* statistical tests between the Phase 1 model (GINE) and its trust-aware

Table 6 Phase 1 supervised performance comparison between traditional ML baselines and GNN models using class-balanced metrics (mean $\pm$ standard deviation across repeated stratified folds)

Model	Sensitivity	Specificity	Macro F1	Macro AUC
Scenario 1: Machine Learning Baselines				
Logistic Regression	0.538 $\pm$ 0.029	0.991 $\pm$ 0.001	0.756 $\pm$ 0.012	0.926 $\pm$ 0.011
SVM	0.393 $\pm$ 0.036	0.996 $\pm$ 0.001	0.744 $\pm$ 0.024	0.954 $\pm$ 0.007
Random Forest	0.030 $\pm$ 0.017	1.000 $\pm$ 0.000	0.524 $\pm$ 0.016	0.954 $\pm$ 0.008
XGBoost	0.513 $\pm$ 0.037	0.997 $\pm$ 0.001	0.806 $\pm$ 0.016	0.960 $\pm$ 0.008
Scenario 3: GNN Models (Phase 1, With Metadata)				
GCN	0.567 $\pm$ 0.204	0.708 $\pm$ 0.084	0.442 $\pm$ 0.025	0.712 $\pm$ 0.036
GAT	0.442 $\pm$ 0.139	0.774 $\pm$ 0.048	0.465 $\pm$ 0.013	0.722 $\pm$ 0.032
GraphSAGE	0.578 $\pm$ 0.189	0.707 $\pm$ 0.067	0.443 $\pm$ 0.019	0.737 $\pm$ 0.055
GINE	0.760 $\pm$ 0.227	0.612 $\pm$ 0.137	0.408 $\pm$ 0.050	0.759 $\pm$ 0.052
GraphSAGE–GAT	0.702 $\pm$ 0.177	0.658 $\pm$ 0.069	0.428 $\pm$ 0.021	0.748 $\pm$ 0.057

Fig. 5 Normalized confusion matrices of traditional machine learning baselines (Logistic Regression, SVM, Random Forest, and XGBoost) in Phase 1. All ML models classify benign lesions almost perfectly, but misclassify a large proportion of malignant samples as benign, resulting in very low sensitivity despite high overall accuracy and specificity

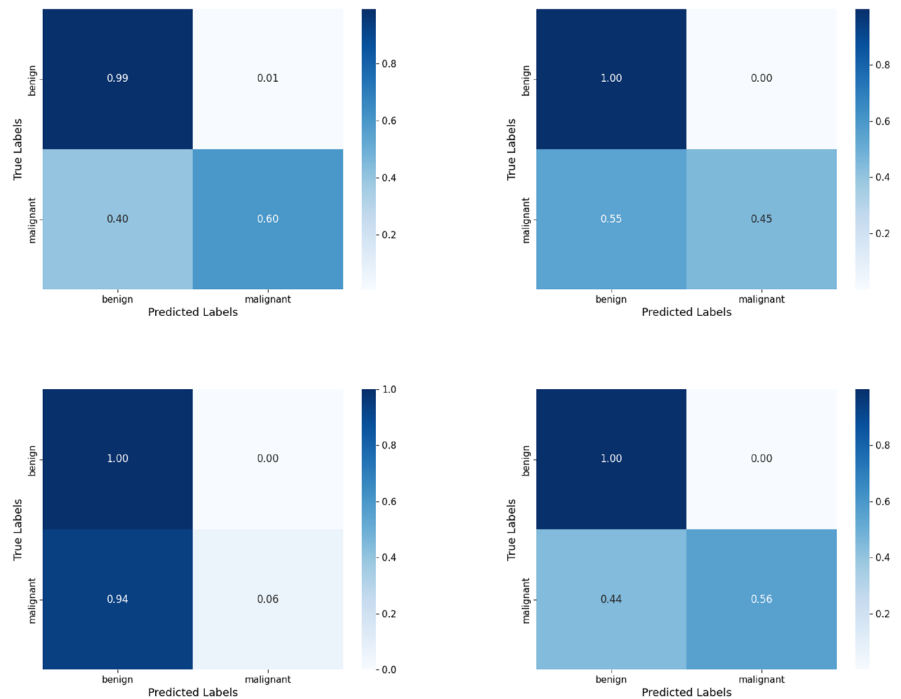


Fig. 6 Normalized confusion matrices of GNN models with metadata-aware graph learning (GCN, GAT, GraphSAGE, GINE, and GraphSAGE-GAT) in Phase 1. Compared to ML baselines, GNN models exhibit more balanced prediction behavior, achieving substantially higher malignant recall while maintaining reasonable specificity

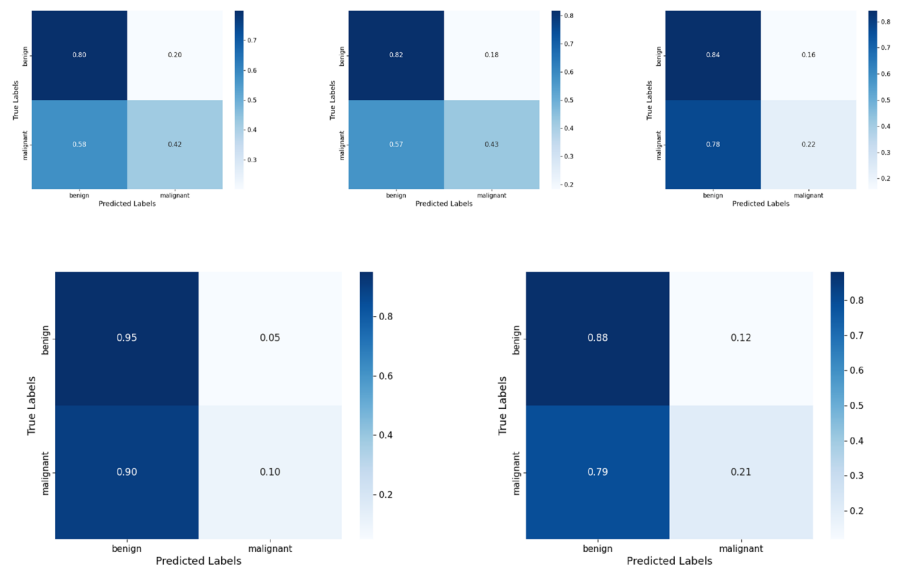


Table 7 Paired *t*-test results comparing Phase 1 (GINE) vs. Phase 2 (GINE_pseudo_finetune) across $n = 15$ repeated stratified folds. Mean Difference is computed as (Phase 2 – Phase 1)

Metric	Mean (P1)	Mean (P2)	Diff (P2–P1)	<i>t</i>	<i>p</i>
Macro F1	0.4015	0.6091	0.2076	17.9804	4.53×10^{-11}
Macro Recall	0.6925	0.8209	0.1283	7.6602	2.26×10^{-6}
Macro AUC	0.7532	0.8853	0.1321	10.5035	5.07×10^{-8}

Table 8 Ablation study on metadata fusion for Phase 1 GNN models. Performance is reported as mean $\pm$ standard deviation across folds. *w/fusion* denotes fusion of deep features, handcrafted features, and metadata (Scenario 3), while *w/o fusion* denotes fusion of deep and handcrafted features only (Scenario 2)

Backbone	Setting	Sensitivity	Specificity	Macro F1	Macro AUC
GAT	w/fusion	0.442 $\pm$ 0.139	0.774 $\pm$ 0.048	0.465 $\pm$ 0.013	0.723 $\pm$ 0.032
GAT	w/o fusion	0.772 $\pm$ 0.046	0.769 $\pm$ 0.029	0.487 $\pm$ 0.012	0.858 $\pm$ 0.010
GraphSAGE	w/fusion	0.578 $\pm$ 0.189	0.707 $\pm$ 0.067	0.443 $\pm$ 0.019	0.737 $\pm$ 0.055
GraphSAGE	w/o fusion	0.763 $\pm$ 0.036	0.786 $\pm$ 0.026	0.495 $\pm$ 0.012	0.862 $\pm$ 0.011
GINE	w/fusion	0.760 $\pm$ 0.228	0.612 $\pm$ 0.137	0.408 $\pm$ 0.050	0.759 $\pm$ 0.052
GINE	w/o fusion	0.785 $\pm$ 0.048	0.760 $\pm$ 0.025	0.483 $\pm$ 0.010	0.855 $\pm$ 0.011
GraphSAGE-GAT	w/fusion	0.702 $\pm$ 0.177	0.658 $\pm$ 0.069	0.428 $\pm$ 0.021	0.748 $\pm$ 0.057
GraphSAGE-GAT	w/o fusion	0.812 $\pm$ 0.034	0.735 $\pm$ 0.028	0.472 $\pm$ 0.012	0.859 $\pm$ 0.010
GCN	w/fusion	0.567 $\pm$ 0.204	0.708 $\pm$ 0.084	0.443 $\pm$ 0.025	0.712 $\pm$ 0.036
GCN	w/o fusion	0.767 $\pm$ 0.043	0.765 $\pm$ 0.025	0.484 $\pm$ 0.011	0.850 $\pm$ 0.009

Fig. 7 Phase 1 training accuracy (left) and training loss (right) of GNN models with metadata fusion (Scenario 3)

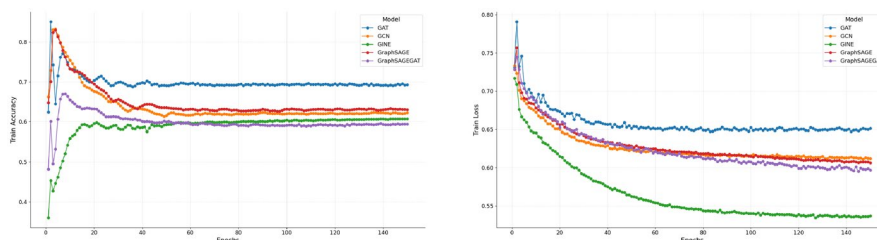
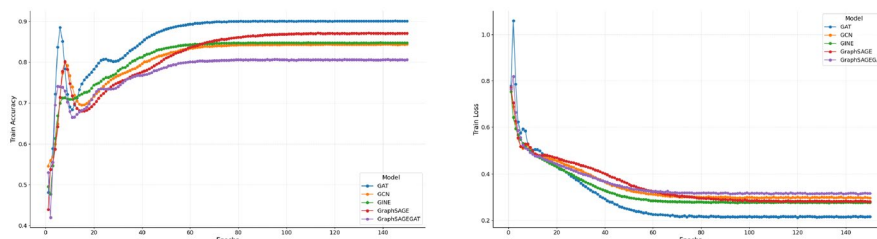


Fig. 8 Phase 1 training accuracy (left) and training loss (right) of GNN models without metadata fusion (Scenario 2)



pseudo-label retrained version (GINE_pseudo_finetune). The paired design is appropriate because both models are evaluated on the same repeated stratified folds, allowing each fold to serve as its own control.

As shown in Table 7, all p -values are far below the standard significance threshold ($p < 0.05$), confirming that Phase 2 yields statistically significant improvements across Macro F1, Macro Recall, and Macro AUC. Notably, the significant increase in Macro Recall supports improved detection of the minority malignant class, while gains in Macro F1 and Macro AUC indicate stronger class-balanced performance and discriminative capability after trust-guided semi-supervised retraining.

4.5.2 Effect of Metadata Fusion

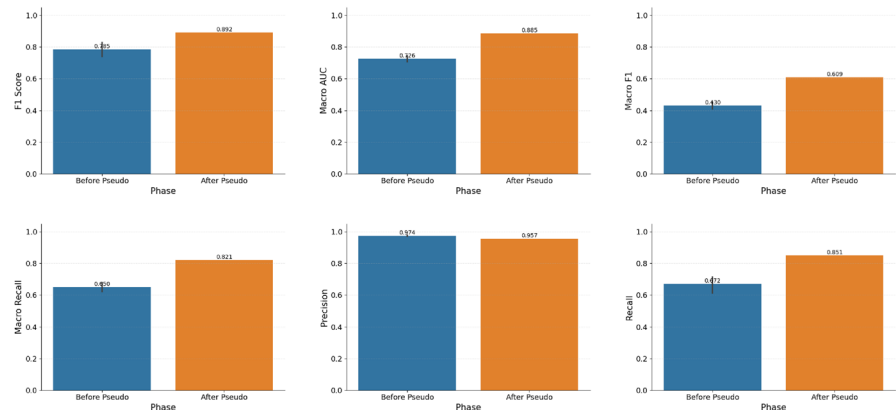
This experiment analyzes the impact of clinical metadata fusion on Phase 1 GNN performance and training behavior. Specifically, we compare two baseline configurations: **Scenario 2 (without metadata)**, where node representations are formed by fusing deep and handcrafted visual features only, and **Scenario 3 (with metadata)**, where visual features are further combined with clinical metadata via metadata-aware attention fusion. All models are trained using identical Phase 1 settings and repeated stratified folds.

From Table 8, purely supervised Phase 1 evaluation shows that Scenario 2 (without metadata) achieves higher Macro AUC and Macro F1 across all backbones. This indicates that when training is limited to labeled data, visual features alone yield more balanced decision boundaries under the current kNN graph construction.

However, training dynamics reveal a different picture. Figures 7 and 8 compare the evolution of training accuracy and loss for all Phase 1 GNN backbones with and without metadata fusion, respectively.

Table 9 Performance of the GINE model before and after Phase 2 pseudo-labeling. Macro metrics denote class-balanced averages

Metric	Phase 1	Phase 2
Accuracy	0.785 ± 0.018	0.892 ± 0.011
Macro AUC	0.726 ± 0.021	0.885 ± 0.014
Macro F1	0.430 ± 0.019	0.609 ± 0.017
Macro Recall	0.650 ± 0.023	0.821 ± 0.016
Precision	0.974 ± 0.006	0.957 ± 0.007
Recall	0.672 ± 0.025	0.851 ± 0.018

Fig. 9 Comparison of classification metrics for the GINE model before (Phase 1) and after (Phase 2) pseudo-label retraining

With metadata fusion, all GNN backbones exhibit smoother convergence and more stable optimization trajectories. In contrast, models trained without metadata tend to plateau earlier and show larger inter-model variability. Among all architectures, **GINE** demonstrates the most consistent and stable behavior in both accuracy and loss curves under metadata fusion, with minimal oscillation across epochs.

Based on this stability and its robust representation learning behavior, GINE with metadata fusion (Scenario 3) is selected as the **Phase 1 backbone** for the subsequent trust-aware semi-supervised training. Although metadata fusion does not directly maximize Phase 1 class-balanced metrics, it provides a more reliable embedding space that is essential for effective pseudo-label refinement in Phase 2 (Scenario 4).

4.5.3 Effect of Phase 2 Pseudo-Labeling

After Phase 1 evaluation, the GINE backbone was selected as the base GNN model due to its stable convergence behavior, smooth optimization dynamics, and consistent class-balanced performance across folds. Phase 2 aims to further improve classification performance by incorporating high-confidence pseudo-labeled samples obtained from the unlabeled test set via entropy-based trust filtering.

Table 9 reports a quantitative comparison between Phase 1 (purely supervised training) and Phase 2 (trust-aware pseudo-label retraining), using metrics computed from `full_model_comparison.csv`. All values are reported as mean $\pm$ standard deviation across folds.

Phase 2 pseudo-labeling yields consistent improvements across all class-balanced metrics. In particular, Macro F1 and Macro AUC increase by $+0.179$ and $+0.159$, respectively, indicating enhanced class discrimination and ranking capability. The slight reduction in Precision reflects a controlled trade-off favoring higher recall and sensitivity for minority malignant cases.

Figure 9 further illustrates the metric-wise improvements achieved after pseudo-label retraining. All macro-level metrics exhibit substantial gains, confirming that the performance improvement is not driven by a single class or evaluation measure.

Fig. 10 Training and validation accuracy (left) and loss (right) of the GINE model across Phase 1 and Phase 2. Phase 2 incorporates trust-aware pseudo-labeled samples

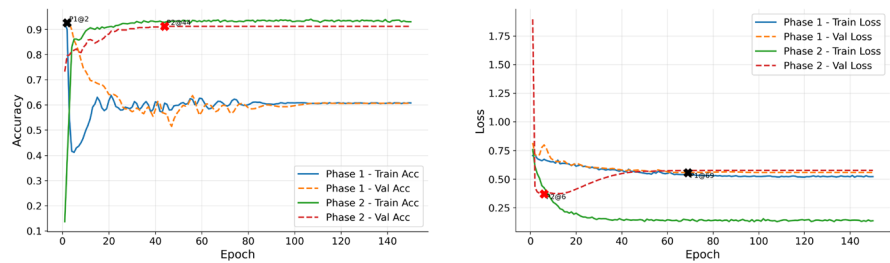
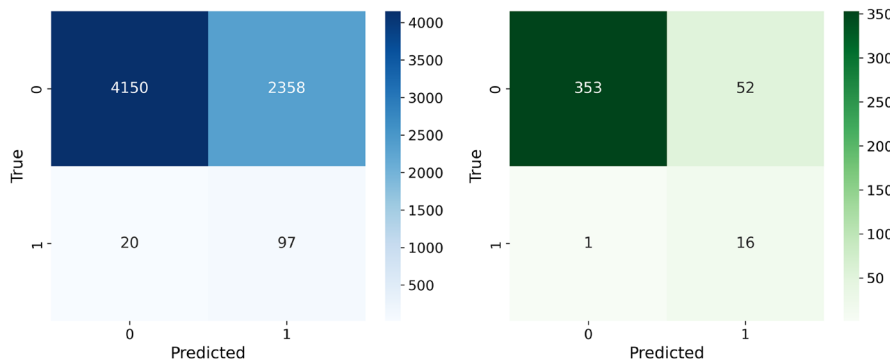


Fig. 11 Confusion matrices of the GINE model before (left) and after (right) Phase 2 pseudo-label retraining. Phase 2 reduces false negatives for the minority class and improves class separation



The effect of Phase 2 pseudo-labeling is further reflected in the training dynamics shown in Fig. 10. While Phase 1 training rapidly saturates at a moderate accuracy level, Phase 2 retraining leads to a stable accuracy increase beyond 0.90 and a pronounced reduction in both training and validation loss.

These results demonstrate that entropy-guided pseudo-labeling substantially enhances the effectiveness of metadata-aware GNN representations. By selectively incorporating high-trust samples, Phase 2 mitigates noisy supervision and enables robust semi-supervised learning beyond purely supervised training.

Figure 11 compares the confusion matrices obtained before and after Phase 2 pseudo-label retraining. The difference in sample counts reflects a change in prediction distribution rather than any modification of the test set.

In Phase 1, the confusion matrix is derived from a purely supervised GINE model. In Phase 2, the same backbone is retrained using entropy-filtered high-trust pseudo-labeled samples, which reshapes the decision boundaries. Although evaluation is performed on the same test split, confidence-aware retraining leads to fewer false negatives and more balanced predictions. No ground-truth test labels are used during pseudo-label generation, ensuring that the observed improvement is not caused by data leakage.

4.5.4 Overall Comparison with State-of-the-Art Methods

To contextualize the performance of the proposed framework, we compare our results with representative state-of-the-art methods reported on the ISIC 2020 dataset, as summarized in Table 1. Most existing approaches primarily rely on image-only deep learning architectures, including CNN ensembles, transformer-based fusion models, or handcrafted-deep feature combinations, without explicitly modeling uncertainty or leveraging unlabeled data.

In contrast, the proposed method integrates multimodal feature fusion, graph-based context modeling, and entropy-guided semi-supervised learning within a unified pipeline. Although direct numerical comparison is not strictly fair due to differences in data splits, preprocessing, and evaluation protocols, our Phase 2 results demonstrate competitive or superior macro-level performance while additionally providing trust-aware prediction control and interpretability. These properties are particularly important for clinical decision support and are largely absent from prior state-of-the-art models.

5 Conclusion and Future Work

This paper presented a trust-aware semi-supervised multi-modal Graph Neural Network (GNN) framework for skin lesion classification on the ISIC 2020 dataset. The proposed approach integrates deep CNN features, handcrafted descriptors, and clinical metadata through a metadata-aware attention fusion module, and leverages entropy-based trust calibration to select high-confidence pseudo-labels for Phase 2 self-retraining. Extensive experiments under repeated stratified cross-validation demonstrate that the proposed framework yields consistent gains in class-balanced metrics after Phase 2, while improving the detection of malignant cases and providing reliability-oriented analysis via trust-level evaluation and confusion-matrix comparisons.

Despite these promising results, several limitations remain. First, clinical metadata in ISIC 2020 contain missing or “unknown” entries (e.g., sex, anatomical site, and age), which may introduce noise and reduce the effectiveness of metadata fusion, especially in purely supervised settings. Second, although prediction entropy provides a simple and effective proxy for uncertainty, it is not necessarily the optimal confidence measure; entropy may be poorly calibrated in highly imbalanced regimes and may not fully reflect epistemic uncertainty, which is critical in safety-sensitive clinical applications.

Future work will address these limitations in two directions. (i) We will investigate calibrated uncertainty estimation to improve trust assignment and pseudo-label filtering, including temperature scaling, calibration-aware training objectives, and Bayesian or ensemble-based uncertainty modeling to better separate reliable from unreliable predictions. (ii) We will explore transformer-based fusion strategies for multimodal representation learning, enabling richer cross-modal interactions than the current attention fusion block and potentially improving robustness under incomplete metadata. These extensions are expected to further enhance both predictive performance and trustworthiness, facilitating safer deployment of semi-supervised multimodal models in real-world dermatological decision support.

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Author Contributions **Thi Trang Nguyen:** Collected and preprocessed the datasets, contributed to feature engineering and assisted in evaluation and visualization.

Van Hieu Vu: Conceived the study, designed the overall architecture, implemented the proposed model, and led the experimental analysis.

Viet Anh Nguyen: Contributed to the development of the GNN framework, supported result interpretation, and co-wrote the manuscript.

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Data Availability The dataset used in this study is publicly available from Kaggle (ISIC 2020): [ISIC2020](#). The source code and experimental pipeline are publicly available at: [Semi-Supervised-GNN-for-Multi-Modal-Skin-Lesion-Classification](#).

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare that they have no competing interests.

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